Ch 11: Inheritance and Polymorphism

Application Program Development

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Introduction to Inheritance

Inheritance

Inheritance allows a new class to extend an existing class. The new class inherits the members of the class it extends.

In the real world, many objects are specialized versions of more general objects

- Example: grasshoppers and bees are specialized types of insect
- In addition to the general insect characteristics, they have unique characteristics:
  - Grasshoppers can jump
  - Bees can sting, make honey, and build hives

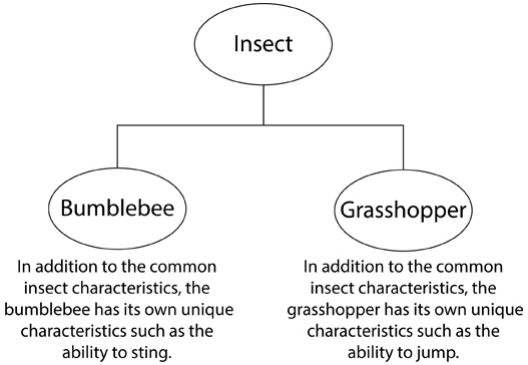


Figure 1: Bumblebees and grasshoppers are specialized versions of an insect (Gaddis, 2021, pg.606)

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Inheritance and the "Is a" Relationship

"is a" relationship

Exists when one object is a specialized version of another object

- Specialized object has all the characteristics of the general object
- Rectangle is a Shape, Daisy is a Flower

For example, need to create classes for cars, trucks and SUVS

- All are automobiles
  - make, year, model, mileage, and price
  - These can be attributes for the base class
- In addition
  - Car has a number of doors
  - Pickup truck has a drive type
  - SUV has a passenger capacity

- **Inheritance**: used to create an "is a" relationship between classes
- **Superclass** (base class, parent class): a (more) general class
- **Subclass** (derived class, child class): a specialized class
  - Extends the superclass
    - Inherits attributes and methods of the superclass
    - New attributes and methods can be added

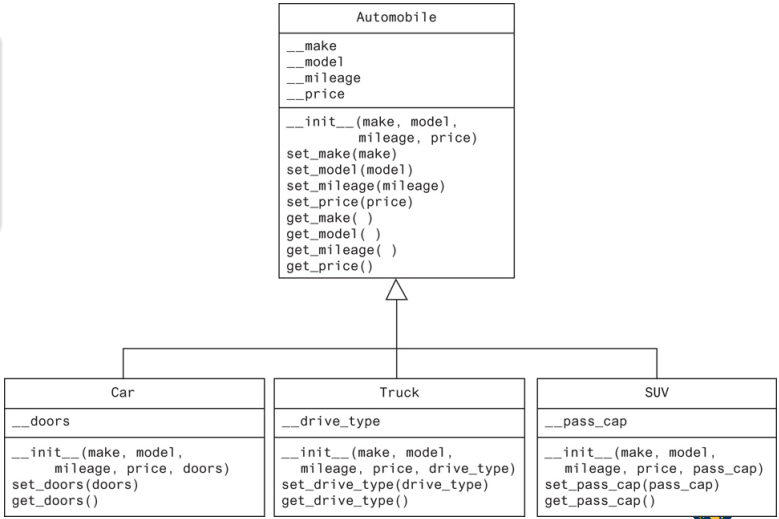
- To indicate inheritance, the superclass name is placed in parentheses after subclass name
  - `class Car(Automobile)`
- The initializer method of a subclass calls the initializer method of the superclass and then initializes unique data attributes
- Add method definitions for unique methods

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Inheritance in UML Diagrams

UML Diagram Inheritance

In UML diagram, show inheritance by drawing a line with an open arrowhead from subclass to superclass



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Polymorphism

- **Polymorphism**: an object's ability to take different forms
- Essential ingredients of polymorphic behavior:
  - Ability to define a method in a superclass and override it in a subclass
    - Subclass defines method with the same name
  - Ability to call the correct version of the overridden method depending on the type of object that called it

- In previous inheritance examples showed how to override the `__init__()` method
  - Called superclass `__init__()` method and then added onto that
- The same can be done for any other method
  - The method can call the superclass equivalent and add to it, or do something completely different



The `isinstance()` Function

- Polymorphism provides great flexibility when designing programs
- **`AttributeError` exception**: raised when a method receives an object which is not an instance of the right class
- **`isinstance()` function**: determines whether object is an instance of a class
  - Format: `isinstance(object, class)`

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Summary

- Inheritance allows a new class to extend an existing class. The new class inherits the members (data and methods) of the class it extends.
- The "is a" relationship exists when one object is a specialized version of another object (inherits all properties of the base class, and extends it).
- Polymorphism is an object's ability to take different forms
  - Overridden methods in subclasses should be invoked if polymorphic



References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

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